

CENTRAL WISCONSIN, WI, USA

WMO: 726465

Lat: 44.783N

Lon: 89.667W

Elev: 389

StdP: 96.74

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94890

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.9	-21.8	-28.9	0.3	-22.8	-26.4	0.4	-21.2	11.1	-9.1	9.9	-8.4	3.9	280	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	30.2	22.4	28.6	21.4	27.3	20.1	23.5	28.4	22.3	26.9	21.1	25.8	4.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	17.6	27.1	20.9	16.4	25.8	19.1	14.6	24.1	72.0	28.8	67.2	26.6	63.1	26.0	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	8.8	7.9	-28.0	32.8	3.5	2.2	-30.5	34.4	-32.6	35.7	-34.6	37.0	-37.2	38.6		
(5)			-27.3	25.6	1.9	1.1	-28.7	26.4	-29.8	27.1	-30.9	27.7	-32.3	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-9.6	-6.3	-1.6	5.9	12.0	17.5	20.3	19.1	15.0	8.0	0.0	-6.3		
	DBStd	11.56	6.16	6.79	6.63	5.45	4.35	4.17	3.25	3.22	4.70	4.82	5.36	6.62		
	HDD10.0	2514	607	457	363	143	28	1	0	0	11	98	301	506		
	HDD18.3	4614	865	690	617	372	202	65	17	29	119	322	551	764		
	CDD10.0	1143	0	0	4	21	89	228	319	283	162	36	1	0		
	CDD18.3	200	0	0	0	0	4	41	78	54	21	2	0	0		
	CDH23.3	1731	0	0	0	14	65	393	674	394	176	14	0	0		
	CDH26.7	407	0	0	0	2	9	94	182	79	39	1	0	0		
(14) Wind	WSAvg	3.6	3.8	3.9	4.0	4.2	3.8	3.1	3.0	2.8	3.2	3.7	3.9	3.6		
(15) Precipitation	PrecAvg	827	28	25	44	73	100	117	93	104	91	72	47	34		
	PrecMax	989	68	50	119	128	159	210	242	328	214	141	144	89		
	PrecMin	656	8	8	7	22	49	46	30	29	6	9	3	8		
	PrecStd	96	16	11	25	28	28	48	49	57	48	32	28	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	10.1	17.9	25.6	27.8	31.2	32.7	31.2	30.9	25.3	15.9	12.0		
		MCWB	4.1	6.6	12.7	16.3	19.2	22.3	24.3	22.9	21.8	18.2	12.2	10.6		
	2%	DB	2.4	6.3	14.4	22.0	25.2	29.0	31.0	28.7	27.5	22.2	12.5	7.1		
		MCWB	0.6	3.1	9.6	13.3	16.4	21.5	23.2	22.0	20.0	16.8	9.4	4.8		
	5%	DB	0.3	3.8	10.9	17.9	22.7	27.5	28.7	27.3	25.3	18.9	9.9	3.5		
		MCWB	-0.6	1.7	7.8	10.3	15.1	20.1	21.8	20.2	18.7	15.0	7.3	1.4		
	10%	DB	-1.2	2.4	7.4	15.5	20.4	26.0	27.3	25.9	22.7	16.2	7.6	1.4		
		MCWB	-1.6	1.0	4.5	8.7	13.6	19.0	20.3	19.4	17.2	12.1	5.6	0.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	7.6	14.0	17.6	20.6	23.5	25.8	24.7	22.3	20.3	13.0	9.9		
		MCDB	6.3	9.4	17.6	24.6	26.0	29.4	32.0	28.7	27.8	24.3	14.5	11.5		
	2%	WB	0.7	3.5	10.8	14.0	18.0	22.3	24.0	22.7	20.8	17.5	10.0	4.6		
		MCDB	2.0	5.4	13.0	19.5	23.6	27.8	28.9	27.2	26.7	20.5	11.6	6.7		
	5%	WB	-0.6	2.2	7.7	11.7	16.3	21.1	22.7	21.3	19.6	15.1	7.5	1.8		
		MCDB	0.3	3.9	9.8	16.4	21.0	26.6	27.1	25.5	24.4	18.6	9.3	3.3		
	10%	WB	-1.8	0.7	4.2	9.5	14.7	19.8	21.4	20.2	18.2	12.8	5.6	0.1		
		MCDB	-1.5	2.4	7.3	14.6	19.2	24.5	25.6	24.3	21.8	15.6	7.8	1.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	9.3	9.8	11.6	11.8	11.5	11.4	11.0	11.5	10.2	7.9	7.9		
		MCDBR	8.8	10.7	13.2	16.2	14.6	13.3	12.8	12.6	13.8	13.8	11.3	10.1		
	5% WB	MCWBR	7.7	8.5	9.2	9.7	8.2	6.8	6.8	6.9	7.7	9.3	8.6	8.2		
		MCWBR	8.6	10.1	12.0	14.8	12.9	12.3	11.5	10.9	12.1	12.7	10.2	9.2		
(40) Clear-Sky Solar Irradiance	taub	taub	0.247	0.277	0.316	0.346	0.380	0.381	0.374	0.375	0.349	0.315	0.285	0.255		
		taud	2.402	2.309	2.366	2.377	2.320	2.346	2.377	2.371	2.448	2.531	2.525	2.452		
	Ebn at Noon	Ebn at Noon	892	914	920	916	889	887	889	875	874	866	839	846		
		Edh at Noon	72	96	105	114	125	122	117	113	97	77	65	62		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.67	2.78	3.92	4.55	5.41	5.91	6.08	5.18	4.09	2.57	1.61	1.20		
	RadStd	RadStd	0.18	0.25	0.41	0.46	0.42	0.45	0.31	0.34	0.29	0.28	0.17	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page